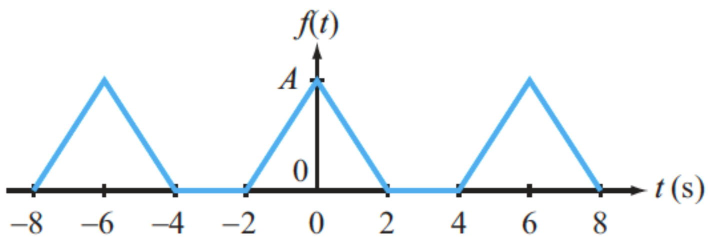
UNIVERSITY of HOUSTON | ECE

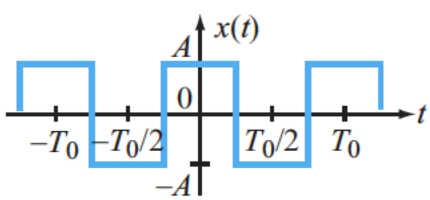
ECE 3337: Signals & Systems Analysis, Class Worksheet 9

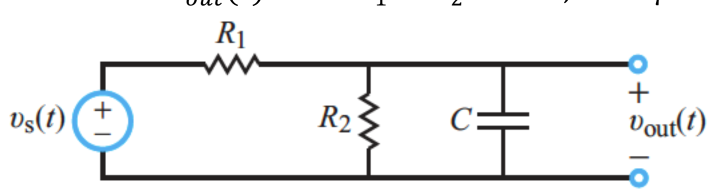
1. (60 points) Calculate the three Fourier series representations for the following waveform, as requested below.



   a. (20 points) Write down the sine/cosine representation of the Fourier series for the above waveform
   b. (20 points) Write down the amplitude/phase representation for the above waveform
   c. (20 points) Write down the exponential representation for the above waveform

2. (40 points) The signal $x(t)$ with A= 10V and T = 1 ms is applied as an input to the following circuit.
   a. (20 points) Derive the Fourier series for the output signal $v_{out}(t)$ taking advantage of table 5-4 of your textbook
   b. (20 points) Calculate the first five terms of $v_{out}(t)$ when $R_1 = R_2 = 2k\Omega, C = 1\mu F$





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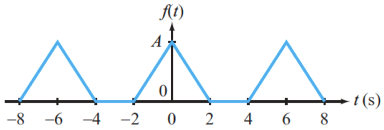

a. (20 points) Write down the sine/cosine representation of the Fourier series for the above waveform
b. (20 points) Write down the amplitude/phase representation for the above waveform
c. (20 points) Write down the exponential representation for the above waveform

$b_n = 0$ Wave form is even

$T = 6 \qquad \omega_0 = \frac{2\pi}{T} = \frac{2\pi}{6} = \frac{\pi}{3}$

$$x(t) = \begin{cases} A - \frac{A}{2}t & 0 < t < 2 \\ 0 & 2 < t \leq 3 \end{cases}$$

$$a_0 = \frac{2}{6}\int_0^2 \left(A - \frac{A}{2}t\right)dt \Rightarrow \frac{2A}{6}\left[2 - \frac{1}{4}(4)\right] = \frac{A}{3}$$

$$a_n = \frac{4}{T_0}\int_0^{T_0/2} x(t)\cos(n\omega_0 t)\,dt = \frac{4}{6}\int_0^2 A(1 - t/2)\cos\left(\frac{n\pi}{3}t\right)dt \Rightarrow \frac{2A}{3}\int_0^2 (1 - t/2)\cos\left(\frac{n\pi}{3}t\right)dt \Rightarrow \left(\frac{2A}{3}\right)\left(\frac{9}{2n^2\pi^2}\right)\left(-\cos\left(\frac{2\pi n}{3}\right) + 1\right) = \frac{3A}{n^2\pi^2}\left[1 - \cos\left(\frac{2\pi n}{3}\right)\right]$$

$$a_n = \frac{3A}{n^2\pi^2}\left[1 - \cos\left(\frac{2\pi n}{3}\right)\right]$$

a) $$\boxed{x(t) = \frac{A}{3} + \sum_{\substack{n=1 \\ n\,odd}}^{\infty} \frac{3A}{n^2\pi^2}\left[1 - \cos\left(\frac{2\pi n}{3}\right)\right]\cos\left(\frac{n\pi t}{3}\right)}$$

b) $$x(t) = C_0 \sum_{\substack{n=1 \\ n\,odd}}^{\infty} C_n \cos(n\omega_0 t + \phi_n)$$

$C_0 = a_0 = \frac{A}{3}, \quad C_n = a_n$

$$\boxed{x(t) = \frac{A}{3} + \sum_{n=1}^{\infty} \left(\frac{3A}{n^2\pi^2}\left(1 - \cos\left(\frac{2n\pi}{3}\right)\right)\cos\left(\frac{n\pi t}{3} + 0^\circ\right)\right)}$$

c.) $x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}, \quad X_n = \frac{C_n}{2} = \frac{3A}{2n^2\pi^2}\left(1 - \cos\left(\frac{2n\pi}{3}\right)\right)$

$$\boxed{x(t) = \sum_{n=-\infty}^{\infty} \frac{3A}{2n^2\pi^2}\left(1 - \cos\left(\frac{2n\pi}{3}\right)\right)e^{\frac{jn\pi t}{3}}}$$

2. (40 points) The signal $x(t)$ with A= 10V and T = 1 ms is applied as an input to the following circuit.
   a. (20 points) Derive the Fourier series for the output signal $v_{out}(t)$ taking advantage of table 5-4 of your textbook
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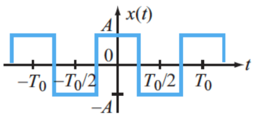
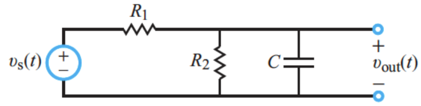
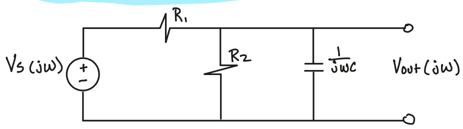




$$V_S(t) = \sum_{n=1}^{\infty} \frac{40}{n\pi} \sin\left(\frac{n\pi}{2}\right) \cos\left(\frac{2n\pi t}{1E-3}\right)$$



$$V_{out} = V_s(j\omega) \frac{R_2 // \frac{1}{j\omega C}}{R_1 + (R_2 // \frac{1}{j\omega C})}$$

$$R_2 // 1/j\omega C = \frac{\frac{R_2}{j\omega C}}{R_2 + 1/j\omega C} = \frac{R_2}{R_2 j\omega C + 1}$$

$$V_{out}(j\omega) = V_s(j\omega) \frac{1}{(RCj\omega) + 2}$$

$$V_{out}(j\omega) = V_s(j\omega) \frac{1}{(j\omega 0.002 + 2)}$$

$$H(j\omega) = \frac{1}{(j\omega 0.002 + 2)}$$

$$V_{out}(t) = \sum_{n\,odd}^{\infty} \left[\frac{40}{n\pi} \sin\left(\frac{n\pi}{2}\right)\right] \cdot |H(nj\,2000)| \cdot \cos(2000n\pi t + \angle H(nj\,2000))$$

$$V_{out}(t) = \sum_{n\,odd}^{\infty} \frac{\frac{40}{n\pi} \sin\left(\frac{n\pi}{2}\right)}{2\sqrt{1 + (2n\pi)^2}} \cos\left(2000\pi n t - \tan^{-1}(2n\pi)\right)$$

Harmonic Analysis — Output Phasors

n	ω (rad/s)	$\lvert V_{in}\rvert$	$\lvert V_{out}\rvert$	φ (°)
1	6283.2	12.7324	1.0006	-80.96°
3	18849.6	4.2441	0.1124	93.04°
5	31415.9	2.5465	0.0405	-88.18°
7	43982.3	1.8189	0.0207	91.30°
9	56548.7	1.4147	0.0125	-88.99°

Circuit Parameters

$R_1 = R_2 = 2$ kΩ

C = 1 μF